

ETL (G01) • Universidad Autónoma de Occidente

Retail Sales Dashboard

Business Requirements Analysis — Lab 1A Pitch

Project Team

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1 The Business Problem

What's happening?

The company has sales, inventory, and product data, but it is scattered across spreadsheets and different systems. Managers cannot get a quick, reliable view of store performance to make timely decisions.

Why does it matter to the organization?

- Reduces time spent searching for information
- Enables data-driven decision-making
- Quickly identifies underperforming products
- Facilitates tracking of goal compliance

Key Stakeholders

- **Regional Managers**
Monitor the performance of their stores
- **Store Managers**
Analyze sales and local trends
- **Marketing Team**
Adjust campaigns based on performance
- **Senior Management**
Obtain strategic indicators

1 Decisions & Missing Information

What decisions could managers make?

Identify stores with low performance

Detect product categories with declining sales

Design promotions for specific products

Compare performance across regions

Evaluate compliance with commercial goals

Plan inventory based on sales trends

Key questions still to resolve

- ? What indicators are priorities for management?
- ? How often do sales goals change?
- ? What access levels will each user have?
- ? How many stores and regions exist?
- ? What system stores the promotions?
- ? What exactly does a "successful store" mean?

2 Business Objectives

1 Centralize sales information into a single analytical view

2 Monitor sales performance by region and store

3 Analyze category and product performance

4 Evaluate compliance with commercial goals

5 Analyze sales trends over time

6 Facilitate clear, easy access to information for managers

7 Analyze the impact of promotions on sales

8 Facilitate operational tracking of transactions

3 Analytical Questions

These questions guide the entire data and dashboard design:

Business Question	Information Required
What is the total sales by day, week, month, and year?	<i>Sales transactions with date and amount</i>
Which categories generate the highest and lowest revenue?	<i>Sales grouped by product category</i>
Which stores have the best and worst performance?	<i>Total sales by store</i>
How do sales vary across regions?	<i>Sales grouped by region</i>
Which stores meet or fail to meet their commercial goals?	<i>Actual sales and goals for each store</i>
What impact do promotions have on sales?	<i>Promotion information related to sales</i>

4 Functional & Non-Functional Requirements

FUNCTIONAL

- Display total sales by day, week, month, and year
- Visualize sales by region and by store
- Display category and product performance
- Filter by region, store, and time period
- Display historical sales trends
- Display goal compliance by store
- Integrate promotion information with sales
- Allow report downloads

NON-FUNCTIONAL

- The dashboard updates information daily
- The platform is easy to use and intuitive
- Only authorized users can access the information
- Indicator loading time under 5 seconds
- Availability during business hours
- Compatible with mobile devices
- Scalable to more stores and data volume
- Consistent and reliable information

4 Data Requirements & Sources

Information Needed	Data Required	Possible Source
Monthly sales	Sales transactions	POS System
Sales by store	Store ID, date, and amount	POS System
Sales by region	Store and region information	POS System + Store Catalog
Product performance	Product catalog and sales	Product Catalog + POS System
Category performance	Categories and sales	Product Catalog
Goal compliance	Commercial goals and actual sales	Commercial Goals System
Active promotions	Campaigns, dates, and products	Promotion System
Historical trends	Sales history	POS System

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User Stories

Regional Manager

I want to **visualize sales of all stores in my region**. So that I can **quickly identify the best and worst performers**.

Store Manager

I want to **view sales by product category**. So that I can **detect which products need promotion**.

Marketing Team

I want to **see which stores meet their commercial goals**. So that I can **adjust advertising campaigns**.

Commercial Director

I want to **download consolidated reports of sales and goal compliance**. So that I can **present them in follow-up meetings**.

Store Manager

I want to **visualize the impact of promotions on sales**. So that I can **identify which campaigns drive the best results**.

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Dashboard KPIs

Total Revenue

Total value of sales made

Sales by Store / Region

Compares performance across branches

Sales by Category / Top 10

Best and worst performing products and categories

Sales Growth (%)

Growth compared to previous periods

Goal Compliance %

Commercial goal achievement

Sales Increase by Promotion

Impact of promotional campaigns

6 Data Integration: The Challenge Behind the KPIs

KPIs that require more than one source

Goal Compliance Percentage

POS System + Commercial Goals System

Sales Increase by Promotion

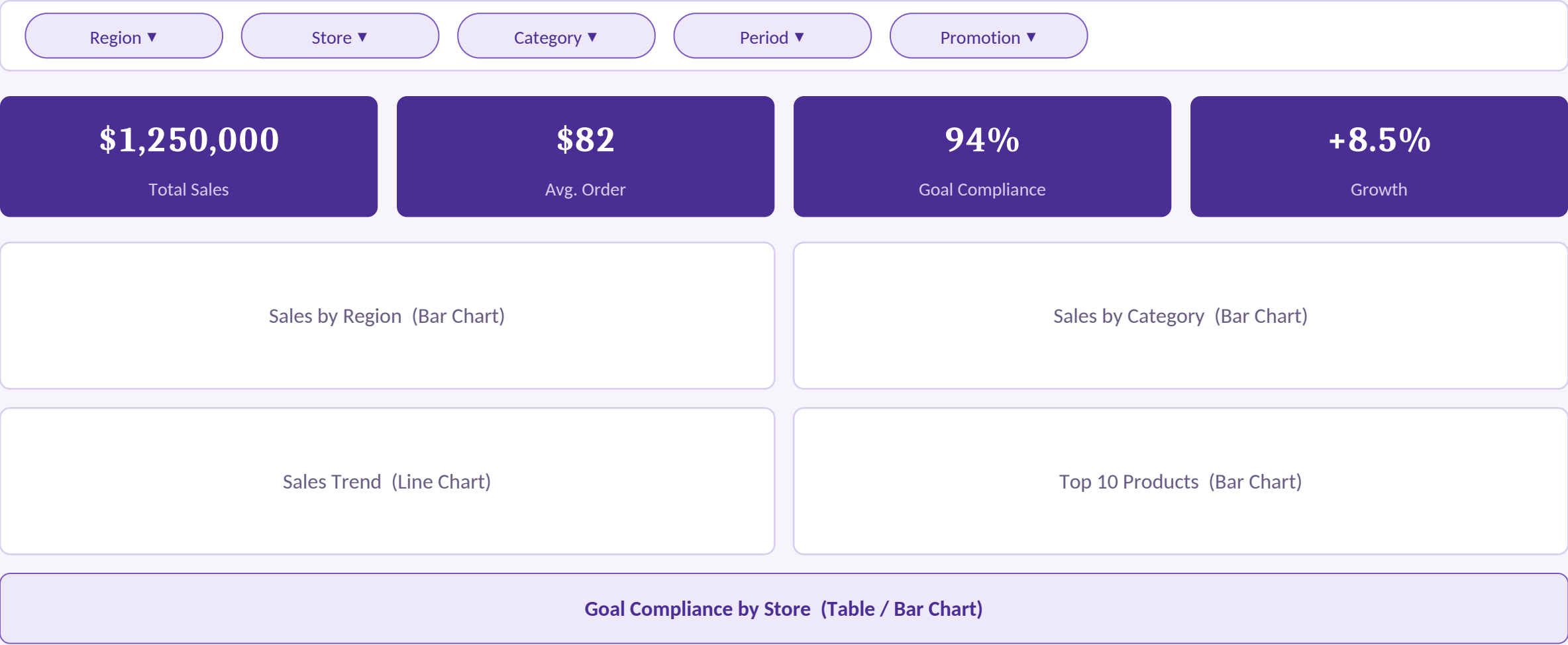
POS System + Promotion System

The Hardest KPI: Sales Increase by Promotion

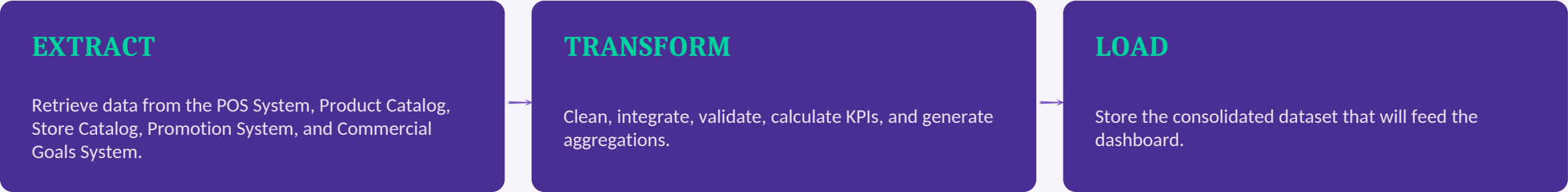
- Integrates sales and promotion data from different systems
- Each promotion must be related to product, store, and date
- Requires comparing sales before, during, and after each campaign
- Attribution risk: must not confuse the effect with seasonality or other simultaneous campaigns

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Dashboard Wireframe



8 From Requirements to ETL (Lab 1B)



Example traceability: requirement → transformation → output

Requirement	Source	Transformation	Expected Output
Measure commercial goal compliance	POS + Commercial Goals System	Integrate actual sales with goals and calculate compliance %	Goal compliance KPI by store and region
Analyze promotion impact	POS + Promotion System	Relate promotions to product, store, and date; compare before/during/after	Promotion impact indicators
Enable report generation	Consolidated ETL dataset	Integrate all processed data and structure it for querying	Final dataset for dashboard and PDF/Excel reports

Final Reflection

Is building a dashboard the first step?

No. Building the dashboard is actually one of the last visible steps, not the first. Before designing any table, chart, or wireframe, the team had to understand the business problem, identify the stakeholders, and translate their needs into objectives and analytical questions.

That work showed there are multiple data sources (POS, Product Catalog, Store Catalog, Commercial Goals System, and Promotion System) that need to be integrated before any KPI can be trusted. The dashboard is the visible outcome of the process — the real first step is understanding the business and the data needed to solve it.

Thank you — questions & discussion